17.2.7 Phylum Mollusca (Soft Bodied Animals)

They are diversified group of animals living in marine, fresh water and terrestrial habitats. Body is bilaterally symmetrical, soft and without segmentation. It is divided into **head, muscular foot** and **visceral mass**. The foot helps in locomotion. The entire body is covered with fold of thin skin called **mantle**, which secretes outer hard **calcareous shell**. Respiration is through gills (**ctenidia**) or lungs or both. Sexes are separate with larval stages during development. e.g-Garden snail, Octopus.

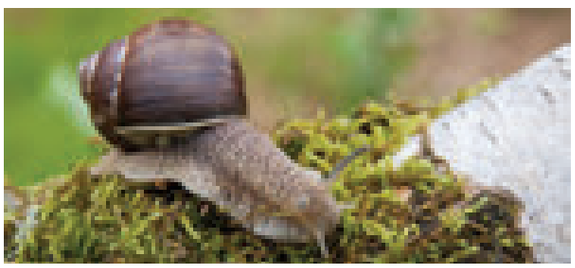


Figure 17.9 Garden Snail



Octopus is the only invertebrate that is capable of emotion, empathy, cognitive function, self awareness, personality and even relationships with humans. Some speculate that without humans, octopus would eventually take our place as the dominate life form on earth.

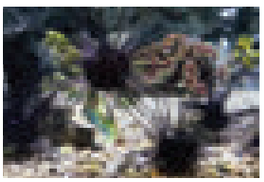


17.2.8 Phylum Echinodermata (Spiny Skinned Animals)

They are exclusively free-living marine animals. These are triploblastic and true coelomates with organ-system grade of organization. **Adult animals** are **radially symmetrical** but **larvae** remain **bilaterally symmetrical**. A unique feature is the presence of fluid filled **water vascular system**. Locomotion occurs by **tube feet**. Body wall is covered with spiny hard **calcareous ossicles**.e.g- Star fish, Sea urchin**.**



Star Fish



Sea Urchin

Figure 17.10 Spiny Skinned Animals

17.2.9 Phylum Hemichordata

Hemichordates are marine organisms with soft, vermiform and unsegmented body. They are bilaterally symmetrical, coelomate animals with **non-chordate** and **chordate features**. They have gill slits but **do not have notochord**. They are **ciliary feeders** and mostly remain as **tubiculous forms**. E.g- *Balanoglossus* (Acorn worms).

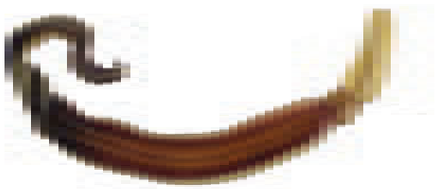


Figure 17.11 Balanoglossus

17.3 Chordata

Chordates are characterized by the **presence of notochord**, **dorsal nerve cord** and **paired gill pouches**. **Notochord** is a long rod like support along the back of the animal separating the gut and nervous tissue. All chordates are triploblastic and coelomate animals. **Phylum Chordata** is divided into two groups: **Prochordata** and **Vertebrata**.

17.3.1 Prochordata

The prochordates are considered as the forerunners of vertebrates. Based on the nature of the notochord, prochordata is classified into subphylum Urochordata and subphylum Cephalochordata.

Subphylum Urochordata

Notochord is present only in the **tail region** of free-living larva. Adults are sessile forms and mostly degenerate. The body is covered with a **tunic** or **test.** e.g. *Ascidian*

IX_SCI_EM_Unit-17_BIO.indd 204 06-02-2020 13:06:05